Latency hints for CFS task

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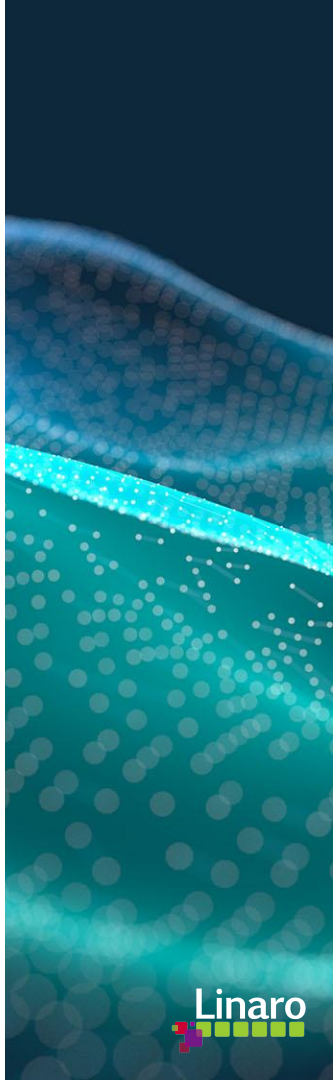
Intro

- CFS (Completely Fair Scheduler) is all about fair share of CPUs
  - Share Compute capacity → Share CPU time
  - Lowest vruntime runs 1st … almost

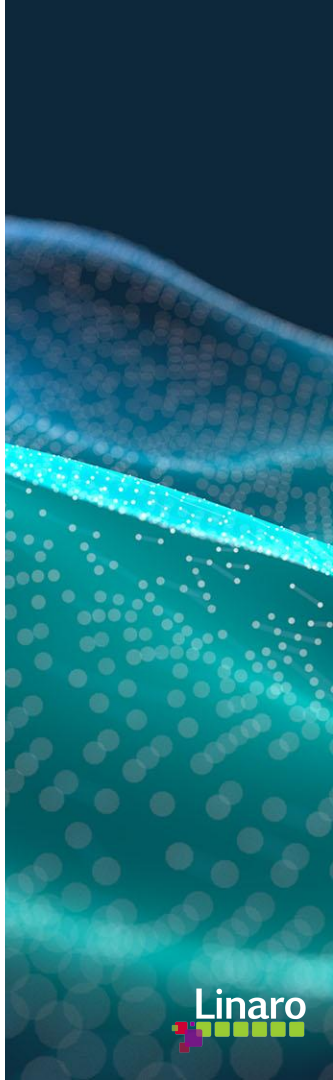
- All CFS tasks don’t have same goal
  - CPU intensive work
  - Background work
  - Feed a media pipeline
  - Part of user interaction path

- What if 2 CFS tasks have same or close vruntime ?
  - Would be good to schedule the latency sensitive task 1st

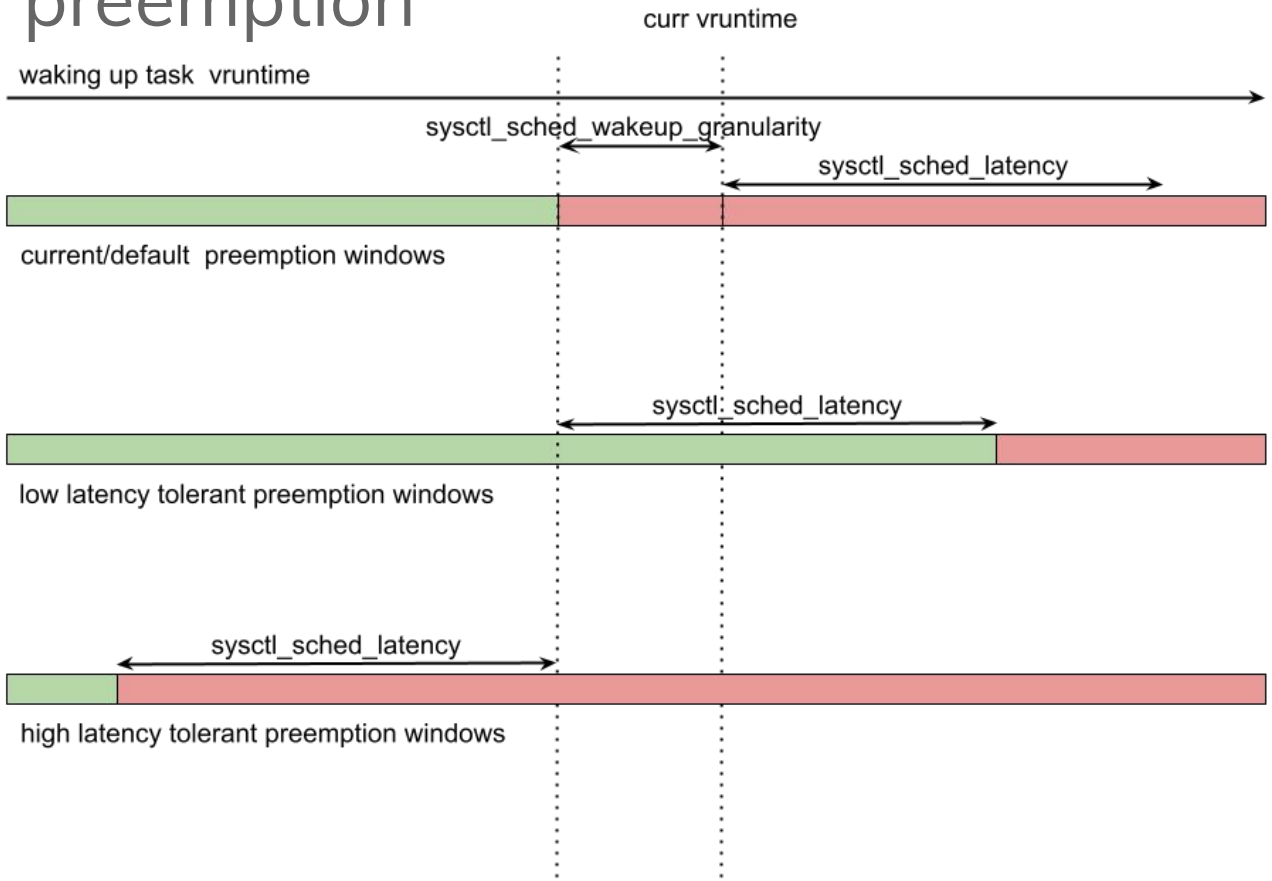


CFS latency

- Add a latency hint to CFS entities
  - Describe its sensitivity to the schedule latency
- Try to schedule the latency sensitive task 1st
- Do not preempt current task by non sensitive to latency tasks
- Do not break fairness

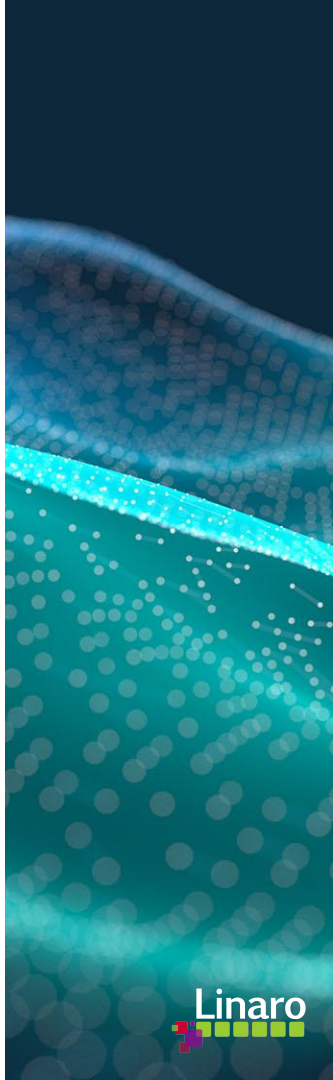


Wakeup preemption



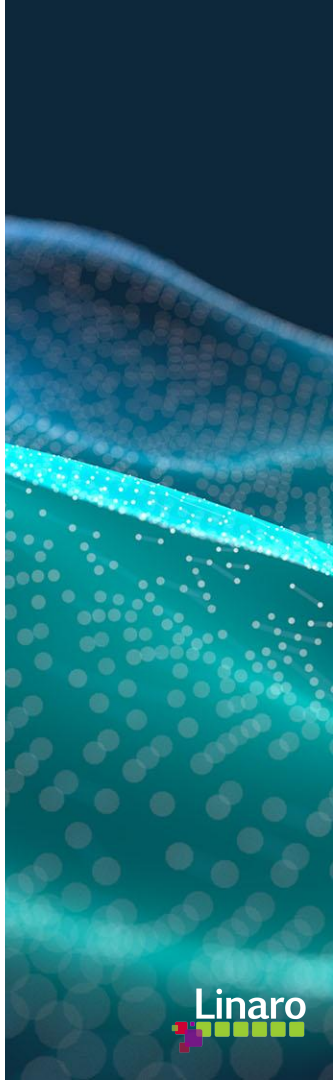
Latency hint

- Comparison done at sched_entity level
  - Include sched group
- Add a latency_offset field
  - Set an offset in the sched_latency ”virtual” period
- Use the offset at wakeup preempt
  - Preempt task with lower vruntime



Processing capacity figures

```
hackbench -p -l (2560 / group) -g group
group
1             1.768(+/- 13%)      0.805(+/- 2%) +54%
4             1.634(+/- 13%)      0.765(+/- 1%) +53%
8             1.305(+/-  4%)      0.745(+/- 2%) +43%
16            0.786(+/-  4%)      0.705(+/- 2%) +10%
```



Scheduling delay figures

```
hackbench -l 10000 -g 16 &
cyclictest --policy other -D 5 -q -n
-H 20000 --histfile data.txt
                 latency 0    latency -20
Min Latencies:      63            63
Avg Latencies:    1459            214
Max Latencies:   58274          82358
50% latencies:     164            87
75% latencies:     848            91
85% latencies:    1246            94
90% latencies:    2149            96
95% latencies:    8463            99
99% latencies:>20000             120
```

```
hackbench -l 10000 -g 16 &
cyclictest --policy other -D 5 -q -n
-H 20000 --histfile data.txt
                 latency 0    latency -20
Min Latencies:      60            61
Avg Latencies:    1077            86
Max Latencies:   87311           444
50% latencies:     92            85
75% latencies:     554           90
85% latencies:    1019            93
90% latencies:    1346            96
95% latencies:    5400           100
99% latencies:   19044           110
```

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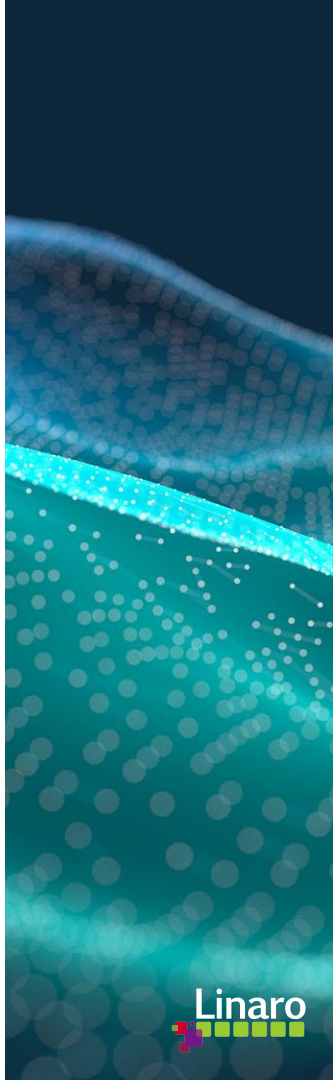
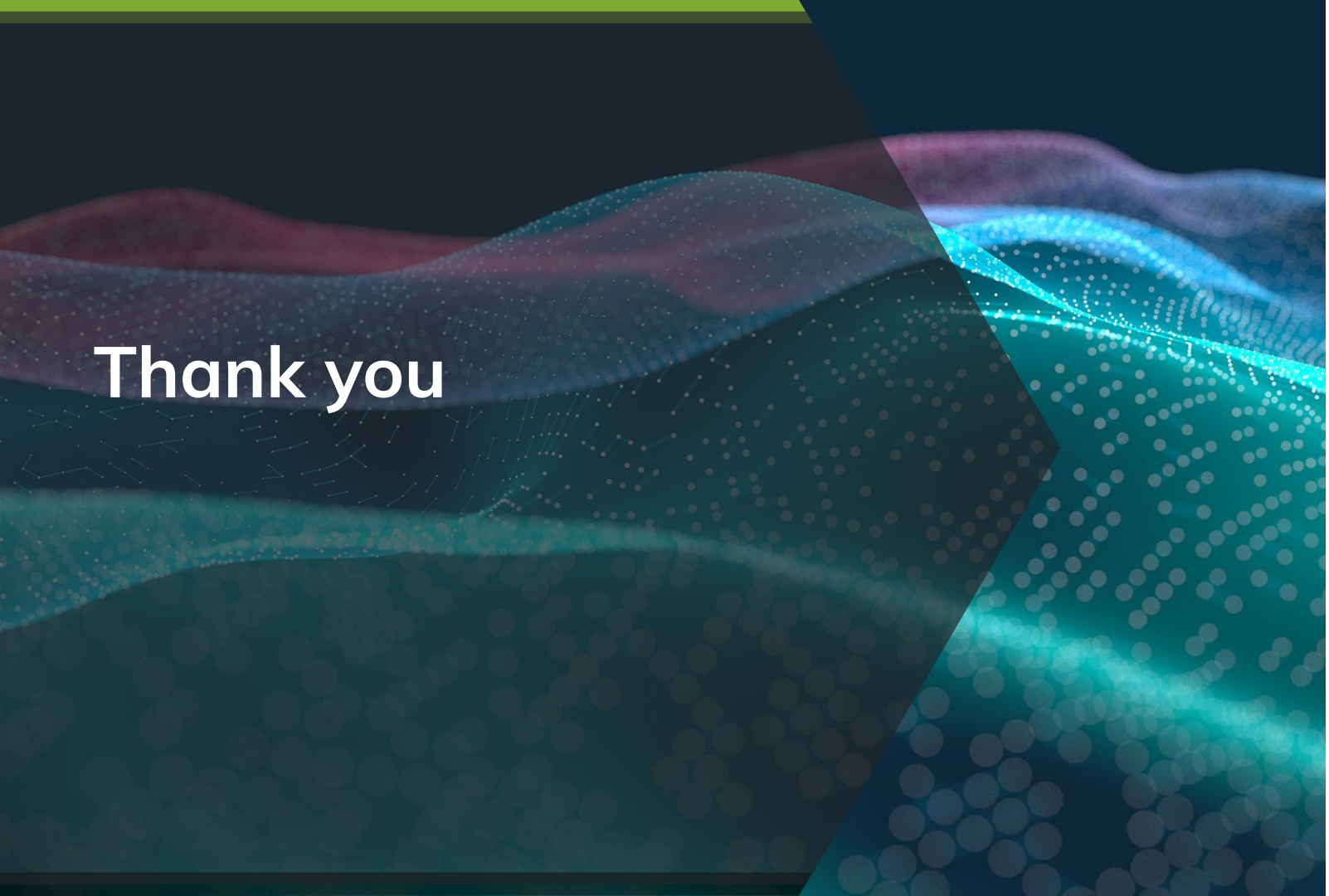
User interface

- Define priority between tasks
- Signed value
  - Negative value : Latency sensitive i.e. not nice with others
  - Positive value: Preemption sensitive i.e. nice with others
- Several attempt to find a generic interface
  - Latency nice prio aligned with nice prio
  - Absolute latency offset value in ns ?
- cgroup interface
  - cpu.latency interface
  - set the latency offset in the range [-sched_latency period:sched_latency]





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